1.1 Definition :

Logic is connected with all types of reasoning. The main aim of logic is to provide rules by using which one can determine the validity of any particular argument or reasoning.

Logic is the branch of mathematics, which deals with the study of reasoning.

1.2 Statement or Proposition :

All declarative sentences whose truth value can be decided is known as statement. All declarative sentences are either true or false. These possibilities are known as truth values.

If the statement is true then we denote it by T or 1 and if the statement is false then it is denoted by F or 0.

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1.3 Types of Statements :

(1) Primary or Simple Statement :

The statement without any logical connectives is known as simple statement or primary statement.

(2) Compound Statement :

The statement with one or more connectives is known as compound statement.

Consider the following sentences.

(i) How are you ? Interrogative Sentence

(ii) What a beautiful girl she is ! exclamatory Sentence

(iii) Ram is god. T

(iv) If Ram is happy and rich then he is educated.

(v) $2 + 2 = 5$. F

It is to be noted that the sentences (i) and (ii) are interogative and exclamatory sentences respectively and it is not possible to decide whether they are true or false and hence they are not propositions or statements. Sentences (iii), (iv) and (v) are declarative sentences whose truth values can be obtained and hence they are statements or propositions. Moreover (iii) and (v) are simple statements as they do not contain any logical connectives and (iv) is a compound statement.

1.4 Logical Connectives (Logical Operators) :

Logical connectives are logical tools required to connect two or more statements.

Connectives :

Connectives	Symbol	Name
NOT	$\sim$ or $\neg$	Negation
AND	$\wedge$	Conjunction
OR	$\vee$	Disjunction
IF - - -THEN	$\rightarrow / \Rightarrow$	Implication
IF AND ONLY IF	$\leftrightarrow / \Leftrightarrow$	Bi-implication

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(1) Negation :

If p is any statement then negation of P is denoted by ~ p or $\neg$p. Moreover if p has a truth value 'T' then ~ p has a truth value 'F' and vice versa.

Truth table

p	~p
T	F
F	T

(2) Conjunction :

If p and q are two statements, then the conjnuction of p and q is denoted by $p \wedge q$ (read as p and q) and is a statement which has a truth value 'T' if both p and q are true, otherwise it has a truth value 'F'.

For example : p : Delhi is the capital of India.

q : London is the capital of Great Britain.

$p \wedge q$: Delhi is the capital of India and London is of great Britain.

Truth table

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Hint : If one is F, then answer is F

(3) Disjunction :

If p and q are two statements, then the disjunction of p and q is denoted by $p \vee q$ (read as p or q) and is a statement which has a truth value 'F' if both p and q are 'F', otherwise it has a truth value 'T'.

If we consider p and q from the above example, then $p \vee q$: Delhi is the capital of India or London is of Great Britain.

Truth table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Hint : If one is T, then answer is T

(4) Conditional or Implication :

If p and q are two statements, then the conditional statement is denoted by $p \rightarrow q$ or $p \Rightarrow q$ (read as if p then q) and has a truth value 'F' if p is 'T' and q is 'F', otherwise it has a truth value 'T' .

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If we consider p and q from the above example, then

$p \rightarrow q$: If Delhi is the capital of India then London is the capital of Great Britain.

Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

(5) Bi-implication or Bi-conditional :

If p and q are two statements, then the Bi-conditional statement is denoted by $p \leftrightarrow q$ or $p \Leftrightarrow q$ (read as p if and only if q) and has a truth value 'T' if both p and q are T and F.

If we consider p and q from the above example, then

$p \leftrightarrow q$: Delhi is the capital of India if and only if London is the capital of Great Britain.

Truth table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Hint:
T = +ve
F = -ve

It is to be noted that

i) $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

ii) $T \wedge F \equiv F$ and $T \vee F \equiv T$

iii) $p \wedge F \equiv F$ and $p \vee F \equiv p$

iv) $T \wedge p \equiv p$ and $T \vee p \equiv T$.

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= T

1.6 Converse , Inverse and Contrapositive :

If p and q are two statements and $p \rightarrow q$ is an implication, then

(i) $q \rightarrow p$ is called its converse (ii) $\sim p \rightarrow \sim q$ is called its inverse

(iii) $\sim q \rightarrow \sim p$ is called its contrapositive.

It is to be noted that every implication is equivalent to its contrapositive.

i.e. $p \rightarrow q \equiv \sim q \rightarrow \sim p$ and $q \rightarrow p \equiv \sim p \rightarrow \sim q$.

We can prove above results by truth table as shown below.

1	2	3	4	5	6	7	8
p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$q \rightarrow p$	$\sim p \rightarrow \sim q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	T	T	T	T	T	T	T

From columns (5) and (8) it is clear that $p \rightarrow q \equiv \sim q \rightarrow \sim p$.

From columns (6) and (7) it is clear that $q \rightarrow p \equiv \sim p \rightarrow \sim q$.

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Example(6) : **Write the converse, inverse and contrapositive of the following statements.**

(1) If a triangle is equilateral, then it is equiangular.

(2) If x is divisible by 2, then it is even.

Solution : (1) p : A triangle is equilateral

q : It is equiangular.

p $\rightarrow$ q : If a triangle is equilateral , then it is equiangular.

Converse : q $\rightarrow$ p : If a triangle is equiangular, then it is equilateral.

Inverse : ~ p $\rightarrow$ ~q : If a triangle is not equilateral, then it is not equiangular.

Contrapositive : ~q $\rightarrow$ ~p : If triangle is not equiangular, then it is not equilateral.

(2) p : x is divisible by 2

q : x is even.

p $\rightarrow$ q : If x is divisible by 2, then it is even.

Converse : q $\rightarrow$ p : If x is even ,then it is divisible by 2.

Inverse : ~p $\rightarrow$ ~q : If x is not divisible by 2 ,then it is not even.

Contrapositive : ~q $\rightarrow$ ~p : If x is not even , then it is not divisible by 2.

1.7 Statement Pattern :

A pattern obtained by combining two or more statements using logical connectives is called statement pattern.

Following are the statement patterns.

(i) Tautology

(ii) Contradiction

(iii) Contingency

(i) Tautology :

In a statement pattern, if all truth values are true then it is called a tautology.

(ii) Contradiction :

In a statement pattern, if all truth values are false then it is called a contradiction.

iii) Contingency :

If a statement pattern is neither a tautology nor a contradiction, then it is called a contingency.

Propositions like :

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Example (7) : Check which of the following are tautology, contradiction and contingency.

(i) $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$

(ii) $((p \vee q) \wedge r) \vee (p \vee q)$

(iii) $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

(iv) $p \wedge (q \wedge \sim p)$

Solution : (i) Given statement pattern is $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$

(i) p	(ii) q	(iii) ~p	(iv) $p \rightarrow q$	(v) $\sim p \vee q$	(vi) (iv) $\leftrightarrow$ v
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

Since all the entries in column (vi) are true, hence given statement pattern is a tautology.

(ii) Given statement pattern is $((p \vee q) \wedge r) \vee (p \vee q)$

(i) p	(ii) q	(iii) r	(iv) $p \vee q$	(v) $(p \vee q) \wedge r$	(vi) $((p \vee q) \wedge r) \vee (p \vee q)$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	T	T	T
T	F	F	T	F	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	F	F	F
F	F	F	F	F	F

Since some entries in column (vi) are true and some are false, hence given statement pattern is a contingency.

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(iii) Given statement pattern is $((p \to q) \wedge (q \to r)) \to (p \to r)$

(i) p	(ii) q	(iii) r	(iv) $p \to q$	(v) $q \to r$	(vi) $p \to r$	(vi) $(p \to q) \wedge (q \to r)$	(vii) $[(p \to q) \wedge (q \to r)] \to (p \to r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	F	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

Since all the entries in the column (vii) are true, hence given statement pattern is a tautology.

(iv) Given statement pattern is $p \wedge (q \wedge \sim p)$

(i) p	(ii) q	(iii) $\sim p$	(iv) $(q \wedge \sim p)$	(v) $p \wedge (q \wedge \sim p)$
T	T	F	F	F
T	F	F	F	F
F	T	T	T	F
F	F	T	F	F

Since all the entries in the column (v) are false, hence the given statement pattern is a contradiction.

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1.8 Logical Equivalence :

Two statements say A and B are said to be logically equivalent if the truth values of A and truth values of B are same.

Example (10) Show that following statements are logically equivalent

(1) $(p \leftrightarrow q) \equiv (\sim p \vee q) \wedge (\sim q \vee p)$

(2) $p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$.

Solution : (1)

i	ii	iii	iv	v	vi	vii	viii
p	q	$\sim p$	$\sim q$	$p \leftrightarrow q$	$\sim q \vee p$	$\sim q \vee p$	(vi) $\wedge$ (vii)
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	F
F	T	T	F	F	T	F	F
F	F	T	T	T	T	T	T

From columns (v) and (viii), we observe that the two given statements are equivalent.

$\therefore p \leftrightarrow q \equiv (\sim p \vee q) \wedge (\sim q \vee p)$

(2)

i	ii	iii	iv	v	vi	vii	viii
p	q	r	$p \wedge r$	$p \rightarrow (q \wedge r)$	$p \rightarrow q$	$p \rightarrow r$	(vi) $\wedge$ (vii)
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	T	F	F	T	F
T	F	F	F	F	F	F	F
F	T	T	F	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

From columns (v) and (viii), we observe that the two given statements are equivalent.

$\therefore p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$

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Example (11) : State and prove De' Morgan's laws.

Soltuion :

First law :

The negation of conjunction of two or more statements is logically equivalent to the disjuction of negation of the statements.

If p and q are two statements, then

$\sim(p \wedge q) \equiv \sim p \vee \sim q$

i	ii	iii	iv	v	vi	vii
p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

From columns (iv) and (vii), we observe that $\sim(p \wedge q) \equiv \sim p \vee \sim q$.

Second law :

The negation of disjunction of two or more statements is logically equivalent to conjuction of negation of the statements.

If p and q are two statements, then

$\sim(p \vee q) \equiv \sim p \wedge \sim q$

i	ii	iii	iv	v	vi	vii
p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim(p \vee q)$	$\sim p \wedge \sim q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

From columns (vi) and (vii) , we observe that $\sim(p \vee q) \equiv \sim p \wedge \sim q$

1.9 Laws of Algebra in Logic :

Propositions satisfy various laws in logic. These laws are useful to simplify expressions. Note that the exception of the innovation law, all laws come in pairs, called dual pairs. For each expression, one finds the dual by replacing all 'T' by 'F' and all 'F' by 'T' and replacing all $\wedge$ by $\vee$ and all $\vee$ by $\wedge$.

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(i) Idempotent laws

(a) $p \vee p \equiv p$

(b) $p \wedge p \equiv p$

(ii) Associative laws

(a) $p \wedge (q \wedge r) = (p \wedge q) \wedge r$

(b) $p \vee (q \vee r) = (p \vee q) \vee r$

(iii) Commutative laws

(a) $p \wedge q \equiv q \wedge p$

(b) $p \vee q = q \vee p$

(iv) Distributive laws

(a) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

(b) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

(v) Identity laws

(a) $p \vee F \equiv p$

(b) $p \wedge T \equiv p$

(c) $T \vee p \equiv T$

(d) $F \wedge p \equiv F$

(vi) Complement laws

(a) $p \vee \sim p \equiv T$

(b) $p \wedge \sim p = F$

(c) $\sim T = F$ & $\sim F = T$

(vii) Absorption laws

(a) $p \wedge (p \vee q) \equiv p$

(b) $p \vee (p \wedge q) = p$

(viii) De-Morgan's laws

(a) $\sim(p \wedge q) \equiv \sim p \vee \sim q$

(b) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

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1.10 Negation of Compound Statements :

(i) $\sim(p \wedge q) \equiv \sim p \vee \sim q$

(ii) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

(iii) $\sim(p \rightarrow q) \equiv p \wedge \sim q$

(iv) $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

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1.11 Arguments and Validity :

An argument is an assertion, that a given set of propositions $S_1, S_2, S_3 - - - S_n$, called premises, yields another proposition S, called conclusion. Such an argument is denoted by $S_1, S_2, S_3 - - - - S_n \rightarrow S$.

An argument $S_1, S_2, S_3 - - S_n \rightarrow S$ is valid if and only if $S_1 \wedge S_2 \wedge S_3 \wedge S_4 - - - S_n \rightarrow S$ is a tautology.

An argument which is not valid is called fallacy.

Note : (i) The following argument is valid

$p, p \rightarrow q \vdash q$ (law of Detachment)

(ii) The following argument is fallacy

$p \rightarrow q, q \vdash p$

(iii) The following argument is valid

$p \rightarrow q, q \rightarrow r \vdash p \rightarrow r$ (law of syllogism).

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For example consider the following argument

S_1 : If Sachin plays cricket, then we will win the match.

S_2 : If we win the match, we will be happy.

$\therefore$ S : If Sachin plays cricket, we will be happy.

Here S_1, S_2 denote the premises of the argument and S is conclusion of the argument.

Let p : Sachin plays cricket

q : We will win the match

r : We will be happy

$\therefore$ $S_1 : p \rightarrow q$

$S_2 : q \rightarrow r$

$S : p \rightarrow r$

Thus $S_1, S_2 \vdash S \Rightarrow p \rightarrow q, q \rightarrow r \vdash p \rightarrow r$.

$\therefore$ By law of syllogism, the given argument is valid.

Example (15) : Prove that the following argument is valid $p \rightarrow \sim q, r \rightarrow q, r \therefore \sim p$.

Solution : Let $S_1 : p \rightarrow \sim q$

$S_2 : r \rightarrow q$

$S_3 : r$

$S : \sim p$

To prove that the above argument is valid, we prove that $S_1 \wedge S_2 \wedge S_3 \rightarrow S$ i.e. $[(p \rightarrow \sim q) \wedge (r \rightarrow q) \wedge r] \rightarrow \sim p$ is a tautology.

1	2	3	4	5	6	7	8	9	10
p	q	r	~p	~q	$p \rightarrow \sim q$	$r \rightarrow q$	$6 \wedge 7$	$8 \wedge 3$	$9 \rightarrow 4$
T	T	T	F	F	F	T	F	F	T
T	T	F	F	F	F	T	F	F	T
T	F	T	F	T	T	F	F	F	T
T	F	F	F	T	T	T	T	F	T
F	T	T	T	F	T	T	T	T	T
F	T	F	T	F	T	T	T	F	T
F	F	T	T	T	T	F	F	F	T
F	F	F	T	T	T	T	F	F	T

Since all the truth values in the column 10 are 'T' , hence the given argument is a tautology and hence valid.

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Example (15) : Test the validity of the following argument.

If I study, then I will not fail in Mathematics.

If I do not play basketball then I will study. **(W-03,07,S-07)**

But I failed in Mathematics.

∴ I must have played basketball.

Solution : Let p : I study

q : I will fail in Mathematics

r : I play basket ball

Then the given argument $S_1 : p \to \sim q$, $S_2 : \sim r \to p$, $S_3 : q$, $\therefore S : r$

$\therefore S_1, S_2, S_3 \vdash S$ i.e. $p \to \sim q, \sim r \to p, q \vdash r$

To prove that the above argument is valid, we show that $S_1 \wedge S_2 \wedge S_3 \to S$

i.e. $[(p \to \sim q) \wedge (\sim r \to p) \wedge \sim q] \to r$ is a tautology.

1	2	3	4	5	6	7	8	9	10
p	q	r	~q	~r	$p \to \sim q$	$\sim r \to p$	$6 \wedge 7$	$8 \wedge 2$	$9 \to 3$
T	T	T	F	F	F	T	F	F	T
T	T	F	F	T	F	T	F	F	T
T	F	T	T	F	T	T	T	F	T
T	F	F	T	T	T	T	T	F	T
F	T	T	F	F	T	T	T	T	T
F	T	F	F	T	T	F	F	F	T
F	F	T	T	F	T	T	T	F	T
F	F	F	T	T	T	F	F	F	T

Since all the truth values in column 10 are 'T', hence the given argument is a tautology and hence valid.

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